Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 9505
CALIBRATION DATE: 28-Feb-19

Slocum Payload CTD CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.012387e+000
h =  1.379359e-001
i = -5.457261e-005
j =  2.247205e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  9.2275e-008
```

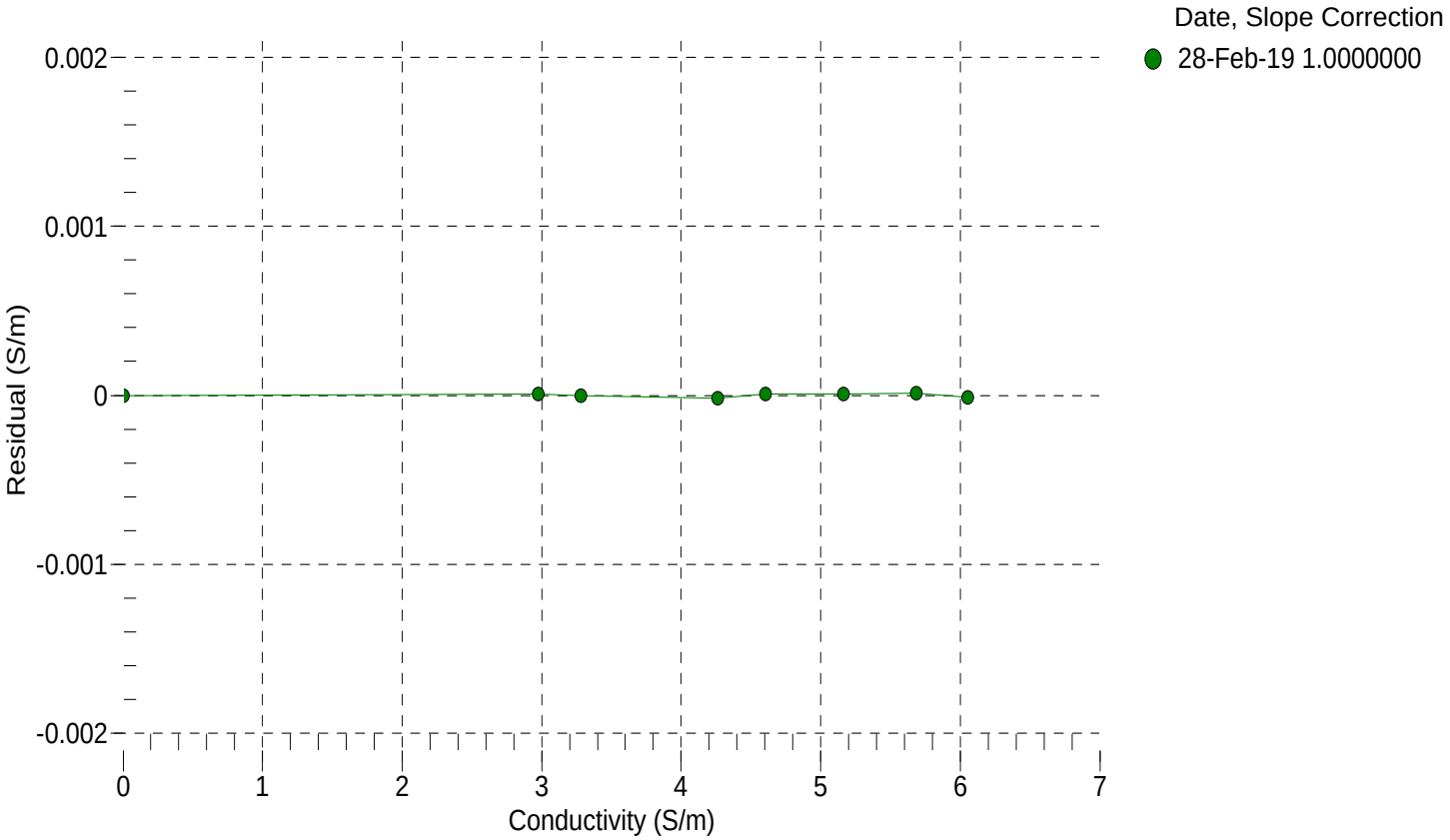
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2708.99	0.00000	0.00000
1.0000	34.7813	2.97326	5368.51	2.97327	0.00001
4.5000	34.7616	3.28007	5570.55	3.28007	-0.00000
15.0000	34.7195	4.26099	6171.60	4.26097	-0.00002
18.5000	34.7101	4.60580	6369.19	4.60581	0.00001
24.0000	34.6995	5.16317	6675.94	5.16318	0.00001
29.0000	34.6922	5.68428	6950.22	5.68429	0.00001
32.5000	34.6849	6.05566	7139.10	6.05565	-0.00001

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 9505
CALIBRATION DATE: 02-Jan-19

Slocum Payload CTD PRESSURE CALIBRATION DATA
1450 psia S/N 11177052

COEFFICIENTS:

```
PA0 =      1.410937e-001
PA1 =      4.511610e-003
PA2 =     -1.166280e-011
PTEMPA0 = -6.706544e+001
PTEMPA1 =  5.113851e-002
PTEMPA2 = -3.900598e-007
```

```
PTCA0 =  5.240441e+005
PTCA1 =  4.708666e-001
PTCA2 =  3.683423e-002
PTCB0 =  2.505637e+001
PTCB1 = -3.250000e-004
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.67	527293.8	1782.4	14.67	0.00
301.76	590916.0	1786.2	301.74	-0.00
588.79	654545.9	1786.9	588.75	-0.00
875.98	718247.2	1787.2	875.99	0.00
1163.12	781946.4	1788.0	1163.13	0.00
1450.21	845646.8	1788.9	1450.18	-0.00
1163.15	781956.5	1788.4	1163.18	0.00
876.03	718262.9	1788.3	876.07	0.00
588.74	654526.2	1787.9	588.67	-0.00
301.76	590932.7	1788.7	301.82	0.00
14.66	527291.9	1789.5	14.66	-0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1977	527357.60
29.00	1906	527359.00
24.00	1806	527338.80
18.50	1695	527316.40
15.00	1625	527328.40
4.50	1414	527316.80
1.00	1345	527302.00

TEMPERATURE (°C)	SPAN
-5.00	25.06
35.00	25.05

y = thermistor output (counts)

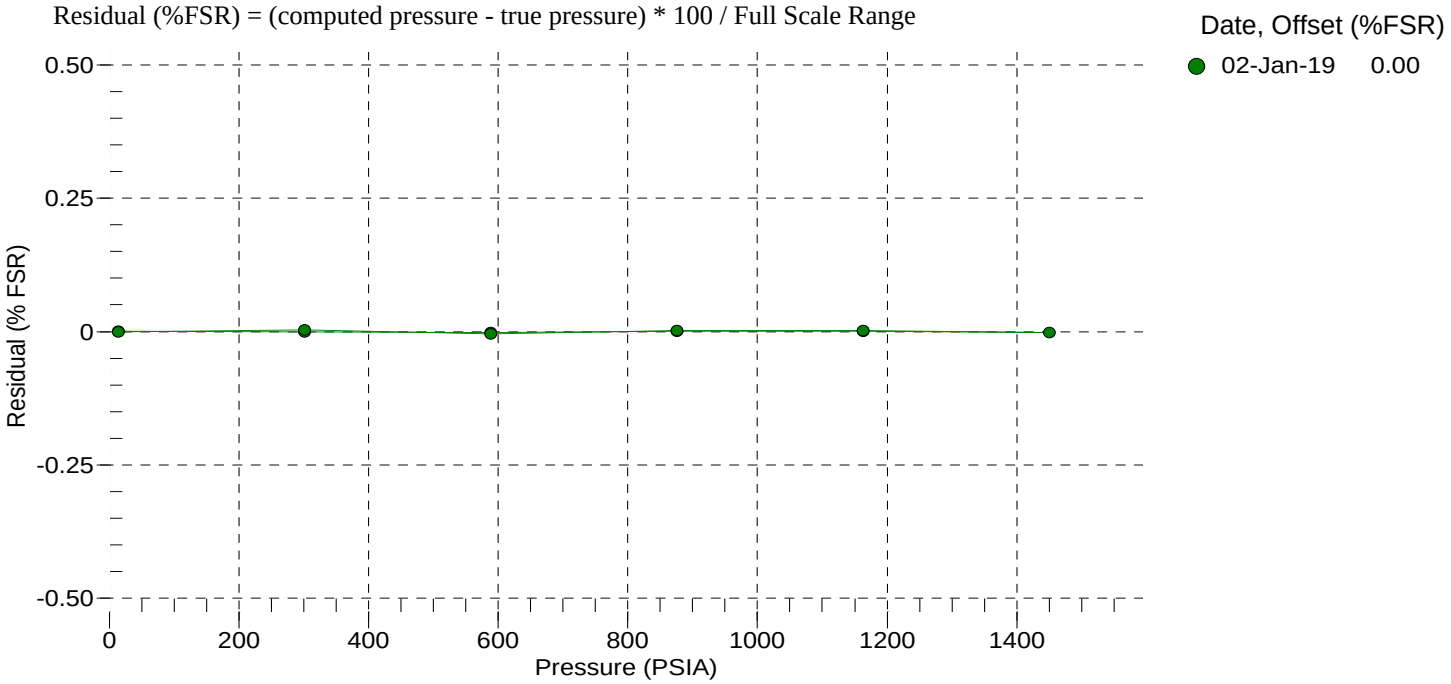
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2019-03-01

Description: Slocum CTD

Sensor Information:

Model Number: Slocum

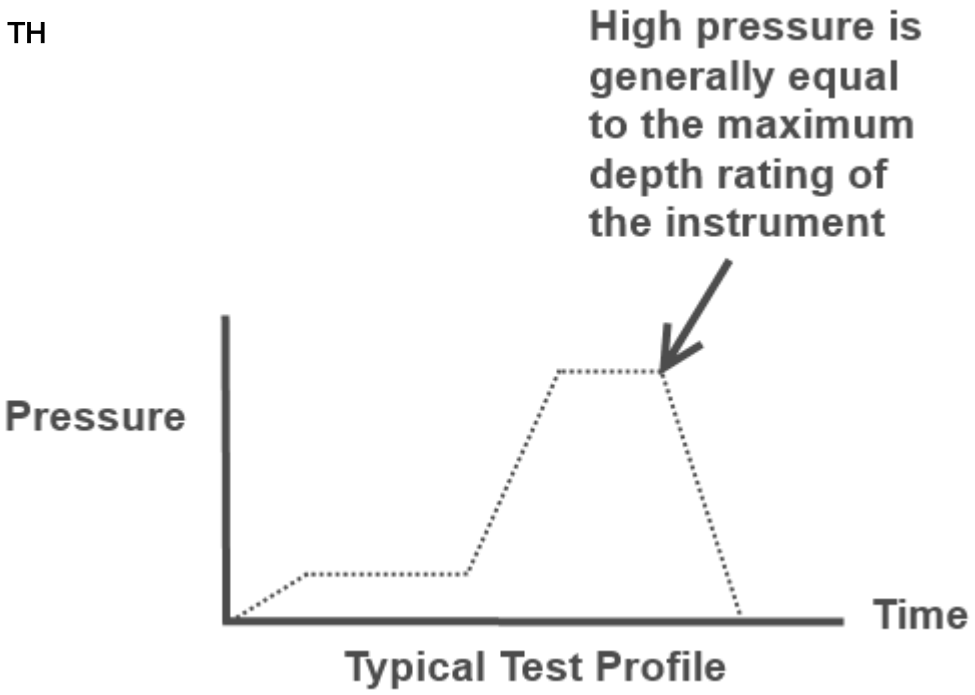
Serial Number: 9505

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	1450	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: TH



Typical Test Profile

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CALIBRATION DATE: 28-Feb-19

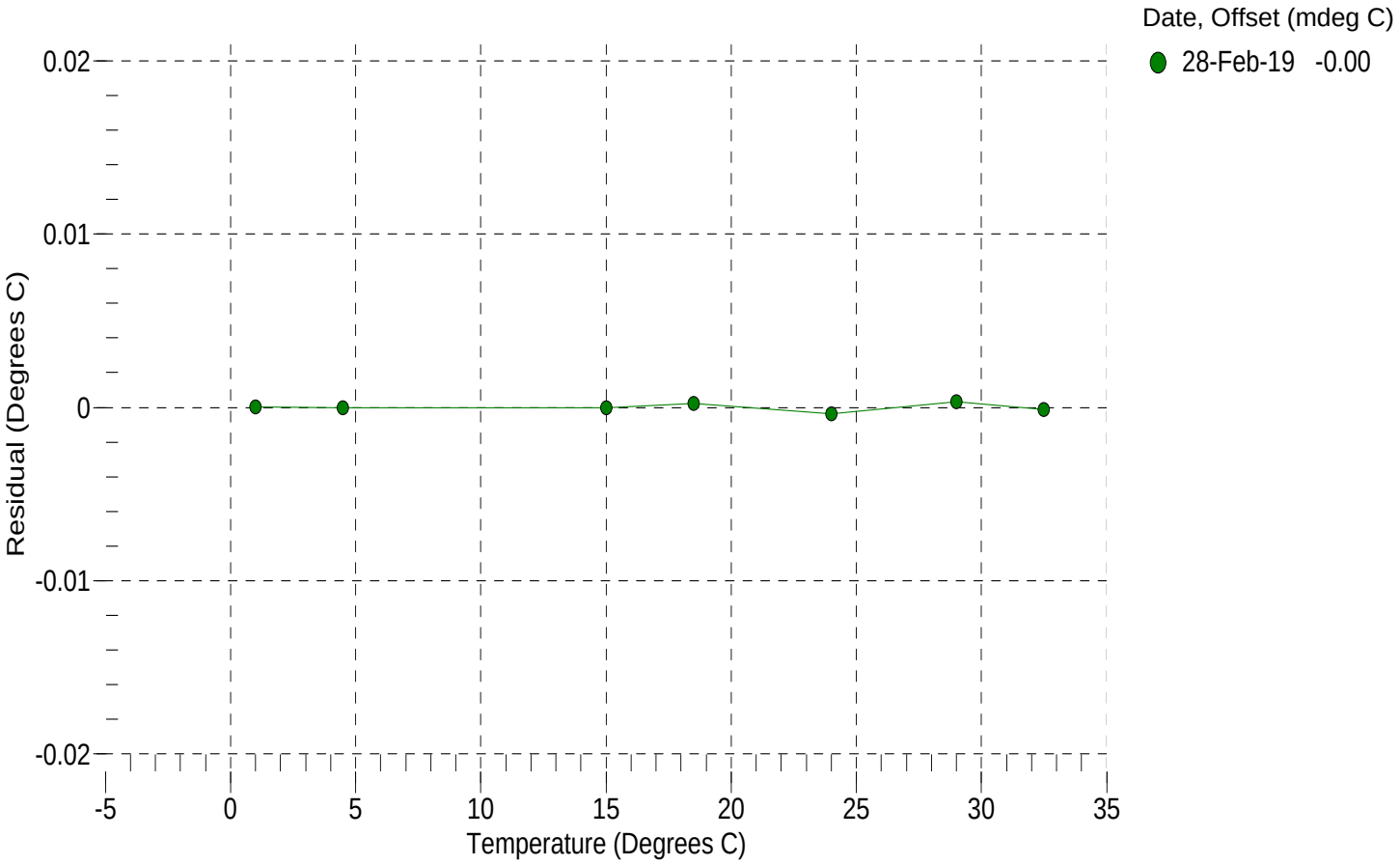
Slocum Payload CTD TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 = -1.509860e-004
a1 =  3.139206e-004
a2 = -4.818932e-006
a3 =  2.093482e-007
```

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	564655.6	1.0000	0.0000
4.5000	483389.6	4.5000	-0.0000
15.0000	309281.4	15.0000	-0.0000
18.5000	268169.6	18.5002	0.0002
24.0000	215615.8	23.9996	-0.0004
29.0000	177907.4	29.0003	0.0003
32.5000	156033.0	32.4999	-0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = 1/{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]} - 273.15

Residual (°C) = instrument temperature - bath temperature